Castles in the Air

Extended chapter commentary

CHAPTER 3
The strange magic of spooky numbers

© John Longley 2023

This version: 2 December 2023

CHAPTER 3 COMMENTARY

The strange magic of spooky numbers

The early part of this chapter's story concerned the expansion of our heroines' ideas on what counts as a 'number': from Whole numbers to Real numbers, and thence to Complex numbers. There is indeed a fascinating story to be told here of the way the very concept of 'number' has developed and expanded over the centuries, which I briefly outlined in the Chapter 3 commentary within the book. In this extended commentary, however, I shall not elaborate further on this story,[a] but instead will concentrate on some of the themes touched on in the latter half of the chapter — especially in Emily's journey with Dirichlet and the reflections it gave rise to.

The big picture

One thing I've tried to get across in parts of the story is a sense of the sheer vastness and scope of mathematics. Mathematics today has become *so* big, in fact, that no given mathematician can hope to understand more than a tiny fraction of it in their lifetime. I don't know of any sensible way to estimate what proportion of today's mathematics I personally 'understand'; but on any reasonable measure, it would be well below 1%.

And mathematics isn't just big, but astonishingly diverse. It's sometimes assumed by outsiders that mathematics is basically all about 'numbers' — but not a bit of it! It's about all manner of shapes, surfaces, waves, networks, symmetries, knots, games, languages, probabilities and much more besides. How does all of this diversity arise?

It's quite common for an area of mathematics to begin with some concept quite familiar from everyday experience — for example, the concept of a triangle, or of a knot, or of a game of chance, or of a network of connected objects. Or it might start from some slightly less obvious concept arising in physics, biology, economics or computing. The first task, often, is to try to distil these concepts into something precise enough to be approached mathematically. But mathematicians very often find that in order to study

a There may be more on this interesting story in a later version of this commentary.

such concepts seriously — to uncover their deeper secrets, as it were — they are led on to consider a whole world of far more abstract and abstruse entities that seem to lie behind or beneath them — things for which one would be hard pressed to find any direct analogy or counterpart in the everyday world. Complex numbers are a good example of this. When we first encounter them, they may seem almost impossible to connect intuitively with any concept of 'number' that we are familiar with. But there are few if any other concepts that are more central to mathematics as a whole than the Complex numbers. (We'll say more in Chapter 5 about how mathematicians reassure themselves that this seemingly fictitious concept makes sense.) Besides the profound role they play in physics, for example, they also appear to hold the key, as Emily learned, to a wealth of information about ordinary Whole numbers.

And this is just the start. Complex numbers were certainly mystifying to many mathematicians when they were invented (discovered?) in the 16th century, but by modern standards, they're not particularly adventurous at all. If it's exotic flavours of 'numbers' you're wanting, there are many other mouthwatering varieties on offer: quaternions, octonions, *p*-adic numbers, surreal numbers. But if you're willing to sample other kinds of abstract entities, the menu is virtually limitless. The higher reaches of mathematics positively teem with strange concepts which one would struggle to explain briefly in everyday terms, frequently going by mystifying names: sheaves, matroids, cobordisms, spinors, orbifolds, martingales. In other words, the simple and accessible concepts that mathematics typically starts from frequently turn out to be just the tip of a much larger iceberg.

All of this can seem pretty daunting. And the curious thing is that it's probably almost as daunting to mathematicians as it is to everyone else. It's not just that I don't *know* what's in 99% of the books in the maths library — I probably wouldn't even be able to *read* the majority of those books, at least not without several weeks or months of preparatory study. Looking at a book on another branch of mathematics feels to me rather like looking at something in a foreign language for which I know a fair bit of grammar but hardly any vocabulary. I think the majority of mathematicians live with this sense that most of the subject is in practice quite inaccessible to them — something that occasionally surprises and bemuses their colleagues in other disciplines.[1]

Cross-connections

I have also tried, in the story, to convey something of the feel of a really big, meaty, complicated proof. Dirichlet's proof of his theorem on primes in arithmetic progressions was one of the high points of 19th-century mathematics, and an epic achievement by the standards of any era. I have based the story version fairly closely on Dirichlet's original paper from 1837,

simplifying and cheating a bit at various points.[2] Don't worry if most of this just washed over you — the main thing I was wanting to get over was just the general sense of a grand argument that combined several different styles of reasoning, but all closely controlled by an overriding master strategy. For readers wanting to understand a bit more of what the mysterious imagery in the story actually refers to, I've written a short supplement called *More on Dirichlet's proof*, available on the book's website.

But the significance of Dirichlet's proof went far beyond the particular result that he proved. For this was the first time that anyone had used methods from 'continuous' mathematics (roughly, the mathematics of the Real or Complex numbers) in order to establish a major new theorem that referred only to Whole numbers. Although certain special cases of Dirichlet's theorem — for instance, Emily's results that there are infinitely many primes of each of the forms $4n+3$ and $4n+1$ — can be proved using the methods of 'basic number theory',[3] [4] Dirichlet found that for his general result, it was necessary to resort to some completely different ideas. In other words, Dirichlet's work busted open a connection between two quite different areas of mathematics which transformed both of them for ever, giving rise to the field known today as *analytic number theory*.

Connections of this sort are of enormous significance, as they allow ideas and methods from one area to be brought to bear on another, leading to fresh discoveries that might never have been made otherwise. So mathematics, far from being a vast collection of separate topics, turns out to be permeated by a deep unity that springs surprises at every turn. By passing through some magic door you may suddenly find yourself transported into an entirely different part of the Castle.

In this way, branches of mathematics which once seemed like outliers have sometimes been brought suddenly centre-stage. A good example concerns the work of Vaughan Jones, a New Zealand born mathematician who worked for much of the 1980s on an obscure problem in infinite-dimensional analysis — a problem, as Alain Connes put it, 'hidden in what was previously thought to be one of the darkest corners, one of the most desert-like regions in all of mathematical geography.'[5] However, a chance meeting with a researcher in topology revealed that Jones's work was exactly what was needed to solve a major problem in the theory of knots. Within a few years, knot theory had been revolutionized, and moreover was finding all sorts of new applications in areas as diverse as quantum theory and molecular biology.

This constantly shifting character of mathematical knowledge — the fact that no one quite knows what connections will pop up next — is part of what makes the subject so intriguing. Just as particular theorems can unexpectedly provide the key to the proofs of other theorems, so an entire branch of mathematics might sometimes turn out to have just what's needed for some other branch of mathematics.

A necessary detour?

Let's return to Dirichlet's theorem for a moment. We've seen that the *statement* of this theorem is relatively simple and involves only Whole numbers, but the proof is very complicated and involves a long detour into what is called *analytic* territory: that of Complex numbers and continuous mathematics. We've also seen that this is an example of a common phenomenon of connections between apparently different topics. But even so, it's not unreasonable to ask whether this long detour is really *necessary*. Might there not be some alternative route that stays strictly within Whole number land? Or do Complex numbers, as Emily wondered, hold secrets about Whole numbers which Whole numbers alone can *never* reveal?

Questions of this kind have an interesting history. For over 100 years after Dirichlet's proof, mathematicians continued to explore and develop the connections between Whole numbers and Complex analysis, obtaining a wealth of new results that couldn't be obtained by any other known means. The most famous of these was the *prime number theorem*: a major result proved in 1896 which gives information about what proportion of Whole numbers are prime up to any given number n.[6] The general consensus at the time was that Complex numbers and continuous mathematics seemed to be essential tools for obtaining these results — that such theorems were simply inaccessible by so-called 'elementary' methods. This was the view expressed by Hardy in 1921:

> No elementary proof of the prime number theorem is known, and one may ask whether it is reasonable to expect one ... A proof of such a theorem, not fundamentally dependent on the theory of [Complex] functions, seems to me extraordinarily unlikely. It is rash to assert that a mathematical theorem *cannot* be proved in a particular way; but one thing seems quite clear. We have certain views about the logic of the theory; we think that some theorems, as we say, "lie deep" and others nearer to the surface. If anyone produces an elementary proof of the prime number theorem, he will show that these views are wrong, that the subject does not hang together in the way we have supposed.[7]

It's important to stress that 'elementary' in this context does *not* mean 'easy', but something a bit more technical, roughly along the lines of 'not depending on ideas from continuous mathematics or Complex number theory'. Whether *this* concept can itself be made precise is another interesting question, which we'll touch on again in Chapter 6.

Hardy's view might have been completely reasonable at the time, but it didn't stand the test of history. In 1946, an 'elementary' proof of Dirichlet's theorem was found by Atle Selberg,[8] and in the following year, 'elementary'

proofs of the prime number theorem were provided by Selberg and Paul Erdős. So in these cases at least, there are routes to the corresponding medallions that *don't* involve long detours through Complex number land. Of course, this doesn't negate the value of the analytic approach: the 'elementary' proofs in question aren't really any *easier* than the 'analytic' ones, and the fact that the analytic proofs were discovered so much earlier is in itself testimony to the value of analytic ways of thinking. Indeed, the analytic approach has frequently enabled number theorists to break new ground, whether or not the same results can, later and with hindsight, be re-proved by more elementary means. And it's also worth noting that the simplest proof of the prime number theorem that we know today — one found in 1980 by Donald J. Newman — is an analytic proof.

But the story doesn't end there. Things have come on a long way since Dirichlet's time, and nowadays, a cutting-edge proof in number theory is likely to involve an array of concepts and theories going far, far beyond the realm of Complex numbers. An iconic example of this is the much-celebrated proof of *Fermat's last theorem*, completed in 1994 by Andrew Wiles after a heroic nine-year struggle.[9] This was a mathematical epic if ever there was one: if Dirichlet's proof seemed to take you round the houses, Wiles's is a trip to the far side of the moon and back. Once again, the theorem itself is just a simple statement about Whole numbers — there are no positive Whole number solutions to the equation $x^n + y^n = z^n$ with $n > 2$ — but the proof relies upon vast tracts of modern mathematics of an entirely different character. Complex numbers are certainly there in the mix, but so too are elliptic curves, modular forms, Galois representations, deformation rings, and a host of other things which are scarcely more than just names to me. At present, no one is entirely certain how much of this machinery is 'essential' to the proof, and how much could in principle be replaced by more 'elementary' arguments, thus avoiding some of the detours through remote and exotic territory. Perhaps this will become clearer in years to come.

The important lesson, then, is that deep results in number theory *may* depend on long detours through other parts of the landscape, and in practice this is frequently how new results are discovered — although it's not always easy to determine how 'necessary' these detours are. In Chapter 8, however, we will see that there are at least *some* kinds of detours, albeit of a rather different character from those mentioned above, which in a certain sense can be *proved* to be necessary: it can be shown definitively that they yield new information about Whole numbers that is not accessible by simpler means.

Deep and shallow theorems

The above quotation from Hardy above also touched on another idea that's worth pondering: the idea that some mathematical results 'lie deeper' than others. Hardy might have been wrong as regards the methods necessary to prove the prime number theorem, but I think that his general idea of theorems of different 'depths' is one that would be instinctively shared by most mathematicians. This is a rather elusive idea and somewhat difficult to convey, though Hardy himself made a good stab at it in his later writing:

> It seems that mathematical ideas are arranged somehow in strata, the ideas in each stratum being linked by a complex of relations both among themselves and with those above and below. The lower the stratum, the deeper (and in general the more difficult) the idea … [It] may happen … that we can recognize and prove, for example, some property of the integers, without any knowledge of the contents of lower strata … But there are also many theorems about integers which we cannot appreciate properly, and still less prove, without digging deeper and considering what happens below.[10]

The depth of an idea has *something* to do with its difficulty, says Hardy, though that's not the whole of it. At present, this concept of 'depth' isn't something which we know how to make precise in most cases, though it's possible that future developments might enable us to do so. But be that as it may, let me try to illustrate this idea by means of a sequence of examples — and along the way, offer you a brief potted history of Fermat's *last theorem*.

We can think of the 'surface' level of mathematics as consisting of specific arithmetical facts that can be checked by calculation, such as $3^2 + 4^2 = 5^2$. This is quite a pretty fact, but we don't really learn much from it: for all we know, it may be just a sort of 'coincidence', a mere curiosity. But next (if you'll allow me a line of algebra) let's consider a more general fact: for any numbers m and n, we have

$$(m^2 - n^2)^2 \quad + \quad (2mn)^2 \quad = \quad (m^2 + n^2)^2$$

This can be checked by school algebra. You will also see that if we set $m = 2$ and $n = 1$, we recover $3^2 + 4^2 = 5^2$ as a special case. What's more, just by plugging in different values for m and n, it's possible to obtain *infinitely many* choices of positive Whole numbers x, y and z that satisfy $x^2 + y^2 = z^2$: the so-called *Pythagorean triples*. So this more general fact goes a bit deeper than the first one: it offers, if you like, a kind of 'explanation' for where $3^2 + 4^2 = 5^2$ comes from, so that this particular example no longer appears as just an isolated 'coincidence', but as part of a general pattern.

Even so, this algebraic equation doesn't tap into anything *very* deep, since its proof requires only the most basic and obvious general properties of addition, subtraction and multiplication. For a theorem that goes a bit deeper still, we'll need to change our problem a little bit.

We've just seen that the equation $x^2 + y^2 = z^2$ has infinitely many positive Whole number solutions. But now let's replace this by $x^4 + y^4 = z^4$. Fermat himself succeeded in proving that this equation has *no* positive Whole number solutions at all (his proof was discovered among his notes following his death in 1665). Now, this certainly *isn't* something that one can show just by shuffling symbols according to the rules of school algebra — we need a logical argument more akin to the other examples of proofs that we've seen. In this case, Fermat's proof is perhaps a bit more demanding than the proof of his 'Little Theorem', but certainly nowhere near as hard as Dirichlet's, and it makes no reference to anything outside the realm of Whole numbers.

What about the equation $x^3 + y^3 = z^3$? Curiously, the case of the exponent 3 turns out to be harder to settle than the case of 4. Once again, it turns out that there are no positive Whole number solutions (this was shown by Euler around 1760), but the proof goes perhaps a little deeper than Fermat's: indeed, it's best to invoke a little bit of Complex number theory to present it in the most natural way.

One can go on like this, showing for specific values of n that there are no positive Whole number solutions to the equation $x^n + y^n = z^n$. The case $n = 5$ was dealt with by Legendre and Dirichlet around 1825; the case $n = 6$ follows immediately from the case $n = 3$; and the case $n = 7$ was solved by Lamé in 1839. But what mathematicians really wanted was some more general method that works for many, or perhaps even all, values of $n > 2$; and for this, one needs to dig deeper still. Fermat himself, famously, had claimed that he had a 'truly wonderful proof' of this; but if he ever did, it never saw the light of day.

The first significant progress on this more general problem had been made by Sophie Germain around 1815. Then in 1850, Ernst Kummer gave a proof that works for many, though not all, values of n by delving into a world of much less familiar structures — rings, ideals, cyclotomic fields — structures belonging to a deeper mathematical stratum which seems to 'explain' what goes on nearer the surface. (Kummer's approach actually deals with every value of n up to 36, but then fails for 37.)

Finally, to cut a very long story short, Andrew Wiles in the 1990s dug sufficiently deep to unearth an enormously powerful result which led to a complete proof of Fermat's last theorem. The deep result concerned yet another unexpected cross-connection between different parts of mathematics: a remarkable relationship between the worlds of *elliptic curves* and *modular forms* which had been conjectured in the 1950s by Goro Shimura and Yutaka Taniyama.[11] This not only provided the key to Fermat's last theorem itself, but shed new light on number theory as a whole. It is very often the case that a

theorem from a very deep stratum provides the key not just to one specific problem, but to a whole class of problems.

There probably aren't very many other *known* results that get deeper than the Shimura-Taniyama theorem. But I would like to add a brief mention of one currently *unknown* result which — assuming it's true at all — is generally agreed to be of quite exceptional depth and power. This is the famous *Riemann hypothesis* — the mysterious citadel which Emily glimpsed briefly from the drawbridge — considered by many to be Number 1 on the Most Wanted List of mathematics today. (It is, incidentally, one of the Clay Institute's list of seven 'Millennium Problems,' each of which carries a $1,000,000 prize for its solution.) It would take me too far afield here to explain what the Riemann hypothesis actually is, but fortunately this has already been done magnificently by Marcus du Sautoy in his book *The Music of the Primes*. The hypothesis has something to do with the same fundamental connection between prime numbers and Complex functions as was exploited by Dirichlet's proof. Although it appears to be enormously difficult to prove, the rewards would be very great: if established, the Riemann hypothesis would give us a much deeper understanding of the seemingly 'random' distribution of the primes, and many other theorems would come within easy reach.[12] This is another splendid example of how a jewel unearthed from very deep down might shed its light across a wide area of mathematics.

Perhaps there are yet deeper treasures buried in strata beyond the likes of Riemann and Shimura-Taniyama. There is, for example, the sweeping vision outlined by the *Langlands programme*, a web of amazing conjectures which promises to provide something like a 'Grand Unified Theory' for large tracts of number theory, algebra and analysis. Realizing a mathematical vision such as this is likely to keep mathematicians busy for many decades to come.

An ultimate theorem?

These grandiose speculations lead one naturally, perhaps, to the question which Emily toyed with as she was drifting off to sleep. Could this succession of deeper and deeper theorems go on forever? Or would one eventually reach something like an Ultimate Theorem that would provide the key to *every* mathematical problem? Let's see if we can clarify what this might mean.

We've seen that a really deep and powerful result often brings a lot of other theorems within easy reach. Even within our story, we saw that Fermat's Little Theorem provided the key to many other discoveries, and the consequences of Shimura-Taniyama or the Riemann hypothesis provide more spectacular illustrations. What would this be like if pushed to its ultimate extreme? An 'Ultimate Theorem', presumably, would in principle bring *every* mathematical theorem within easy reach — *so* easy, perhaps, that solving any other problem

might then become just a matter of routine calculation. Pushing this idea to its limit, we might even fantasize about a purely mechanical method that achieved this: a Master Engine into which you could feed any mathematical problem you wanted, turn the handle, and eventually obtain the answer. Needless to say, we don't actually know of any such algorithm. But *could* it exist? Is it worth searching for one? Or is the idea too crazy to take seriously?

In a remarkable series of lectures and writings from the 1920s, David Hilbert, a mathematician of exceptional breadth and depth and viewed by many as the king of mathematics in his day, dared to suggest that a Master Algorithm along these lines might really be possible. What's more, he and his colleague Wilhelm Ackermann claimed to have made some significant progress towards discovering such an algorithm, and it appears that Hilbert fully expected that it was only a matter of time before it was found.

To understand more exactly what Hilbert and Ackermann were proposing, we will need a few further concepts, which we shall introduce over the next few chapters. For the time being, though, suffice it to say that they really were, in effect, proposing a procedure which, if it existed, would *in principle* be able to deliver straight Yes-No answers to a huge class of mathematical questions — including (for example) Fermat's last theorem, the Goldbach conjecture and the Riemann hypothesis — after a suitably enormous amount of purely handle-cranking, mechanical calculation. The challenge of finding such a master algorithm was termed the *decision problem* (often referred to by its rather portentous German name, the *Entscheidungsproblem*).[13]

Many mathematicians reacted with understandable scepticism to this idea. For one thing, experience suggested that there was no limit to the potential difficulty of mathematical problems, and that human ingenuity and cunning played an essential role in discovering proofs. But on a more emotional level, perhaps, many of them didn't *want* Hilbert to be right, since they didn't like the idea that all the creativity, imagination and genius they poured into their proofs could be supplanted by mindless symbol-crunching. The great mathematician and physicist John von Neumann wrote:

> The undecidability [*i.e.* the absence of a general decision algorithm] is even the necessary condition for the contemporary practice of mathematics … to make sense. The very day on which the undecidability would cease to exist, so would mathematics as we now understand it; it would be replaced by an absolutely mechanical prescription.[14]

Hardy was even more outspoken:

> There is of course no such theorem and this is very fortunate, since if there were we should have a mechanical set of rules for the solution of

> all mathematical problems, and our activities as mathematicians would come to an end … it is only the very unsophisticated outsider who imagines that mathematicians make discoveries by turning the handle of some miraculous machine.[15]

Whilst I can perfectly well sympathize with these sentiments, I can't help being reminded here of an episode from Douglas Adams's *The Hitchhiker's Guide to the Galaxy*, in which two representatives from the Philosophers' Union try to hijack a ambitious project to calculate the Ultimate Answer to Life, the Universe and Everything, on the grounds that it would put philosophers out of work if it succeeded. To be fair, Hilbert and Ackermann hadn't claimed that their algorithm would be usable *in practice* — they were fully aware that the required computations would probably be infeasibly long in most cases — but perhaps even the prospect of being dispensable *in theory* was more than many mathematicians could accept.

Despite the scepticism of many, it's undeniable that Hilbert's idea created quite a buzz. Many were intrigued by the idea, even if ultimately unconvinced. Hilbert and Ackermann themselves modestly averred that 'the *Entscheidungsproblem* should be considered the main problem of mathematical logic,'[16] a verdict that was echoed by many of their contemporaries. Jacques Herbrand wrote in 1930 that

> In a sense it is the most general problem of mathematics … Although at present it seems unlikely that the decision problem can be solved, it has not yet been proved that it is impossible to do so.[17]

What became of Hilbert's extraordinary idea? Is such a mathematical Holy Grail attainable? We shall discover answers to these questions within the next two chapters, in which we leave behind our survey of mathematics in general and enter the domain of mathematical logic proper.

Endnotes

1. I recall a BBC radio conversation in the 1980s between the mathematician Christopher Zeeman and the biologist Lewis Wolpert. Zeeman's remark that "most of mathematics is a closed book to me" drew considerable astonishment from Wolpert.
2. An English translation of Dirichlet's original paper, *There are infinitely many prime numbers in all arithmetic progressions with first term and difference coprime*, can be located on the Internet. For a wealth of biographical information on Dirichlet's life and work, see J. Elstrodt, *The life and work of Gustav Lejeune Dirichlet*, in Duke, W. and Tschinkel, Y. (eds.), *Analytic number theory: A tribute to Gauss and Dirichlet*, Clay Mathematics Institute, pp. 1-37, 2007.

3. The argument for odious primes goes like this. Suppose there are only finitely many odious primes. Multiply them together to get a big number N. Now N is an odd number (being a product of odd numbers), so let's say $N = 2n + 1$. Now consider the number $M = 2N + 1$ (rather than just $N + 1$ as in Euclid's proof). Then $M = 4n + 3$, that is, M is odious. Now let's look at the prime factors of M. They can't all be evil, because a bit of algebra shows that a product of evil numbers is always evil:

$$(4a + 1)(4b + 1) = 16ab + 4a + 4b + 1 = 4\,(4ab + a + b) + 1$$

So at least one prime factor of M (call it p) has to be odious. Now p divides $M = 2N + 1$, so p can't divide $2N$, and in particular p doesn't divide N. But this is a contradiction, since N is supposed to be the product of all odious primes. So our original supposition must have been wrong: there must be infinitely many odious primes after all.

You might now wonder whether we can show in the same way that there are infinitely many *evil* primes, just by swapping 'evil' and 'odious' everywhere in the above proof. Unfortunately, this hits a snag, because it's *not* true that a product of odious numbers is always odious (for example, $3\times3 = 9$).

4. The case of evil primes turns out to be harder. The result was first stated (without proof) by Euler in 1775. Several proofs are known; the one I've attributed to Emily was found by V.-A. Lebesgue in 1856. It requires some modulo arithmetic, and uses Fermat's Little Theorem.

Suppose there are only finitely many evil primes. As in the last proof, multiply them together to get a big number N. Now N is odd (being a product of odd numbers), so let's say $N = 2n + 1$.

Now for the clever part. Let $M = N^2 + 1$. Then $M = 4(n^2 + n) + 2$, so $M > 4$ but M is not divisible by 4. In particular, M is not a power of 2. But this means that M has at least one *odd* prime factor, say p. Since p divides $N^2 + 1$, we have that $N^2 \equiv -1 \pmod{p}$.

We will show next that p must be evil. To do this, let's suppose p were odious (say $p = 4a + 3$), and try to obtain a contradiction. Under this assumption, we'd have

$$\begin{aligned} N^p &= N^{4a+3} = (N^2)^{2a+1} \times N \equiv -1^{2a+1} \times N \\ &\equiv -1 \times N \equiv -N \pmod{p} \end{aligned}$$

On the other hand, Fermat's Little Theorem tells us that $N^p \equiv N \pmod{p}$. So we must have $N \equiv -N \pmod{p}$. Adding N to both sides, we get $2N \equiv 0 \pmod{p}$, or in other words, p divides $2N$. But this is impossible if p is odious: we defined N to be the product of all evil primes, so $2N$ has no odious prime factors. This gives the contradiction we were looking for, and we can conclude that p must be an evil prime after all.

We're now nearly done. On the other hand, we've said that p divides $N^2 + 1$, so p doesn't divide N^2, and so p can't divide N either. But this contradicts the

fact that N is the product of all evil primes. So there must be infinitely many evil primes after all.

Incidentally, the terms 'evil' and 'odious' as used here are *not* standard mathematical terminology!

5. In Changeux and Connes (1995), p.49.
6. The prime number theorem says that the number of primes up to a given Whole number n is quite close to n / ln(n), where ln(n) denotes the logarithm of n to the base e = 2.7182812 … More precisely, the theorem says that the *ratio* between the true number of primes up to n and the value given by the formula n / ln(n) converges on 1 as n gets larger and larger.

   In a way, the prime number theorem is a better example than Dirichlet's of the use of Complex analysis in number theory, since its proof goes deeper into the peculiar and distinctive magic of Complex numbers than Dirichlet's does. The disadvantage is that the theorem is more complicated to state, and certainly not easy to present as a statement purely about Whole numbers in the way that Dirichlet's theorem obviously is.
7. In G.H. Hardy, Copenhagen lecture, 1921; quoted by H. Bohr in *Address of Professor Harald Bohr*, Proceedings of the International Congress of Mathematicians vol. 1, Cambridge, 1950.
8. See A. Selberg, *An elementary proof of Dirichlet's theorem about primes in an arithmetic progression*, Annals of Mathematics 50(2), 1949. Selberg explains that his proof is 'more elementary in the respect that we do not use the complex characters mod k, and also in that we consider only finite sums.'
9. For a popular account of this moving human story, with some maths woven in, see Singh (1997).
10. In Hardy (1940), §17.
11. Actually, Wiles didn't prove the Shimura-Taniyama conjecture in its fully general form, but only for so-called *semistable* elliptic curves, which was enough for a proof of Fermat's Last Theorem. The full conjecture was established in 1999 by Breuil, Conrad, Diamond and Taylor, extending Wiles's work.
12. For example, some of the clever tests for recognizing prime numbers which we alluded to earlier in the chapter, such as the Miller-Rabin primality test, would be known definitively to work in all cases if (a slightly generalized version of) the Riemann hypothesis were true.
13. See Hilbert and Ackermann (1928), Chapter 3. The authors are explicit that a solution to the *Entscheidungsproblem* would yield a decision algorithm for branches of mathematics that could be based on finitely many (first-order) axioms — which, as Hilbert realized, would suffice for settling problems such as Fermat's last theorem (see *e.g.* Hilbert (1927)). The class of statements that could be treated in this way would include the class of *Predictive* statements whch we shall meet in Chapter 5.
14. In J. von Neumann, *Zur Hilbertschen Beweistheorie*, Mathematische Zeitschrift 26, p.11-12, 1927.
15. In G.H. Hardy, *Mathematical proof*, Mind 38(149), p.16,18, 1929.
16. In Hilbert and Ackermann (1928), p.77.

17. In J. Herbrand, *Sur le problème fondamentale de la logique mathématique*, 1931, English translation in Herbrand (1971), pp.215-259. For further historical background on the decision problem, including quotations from other logicians of the period, see the introduction to Börger, E., Grädel, E. and Gurevich, Y., *The classical decision problem*, Springer, Berlin, 1997.